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EXPERIMENT – 18

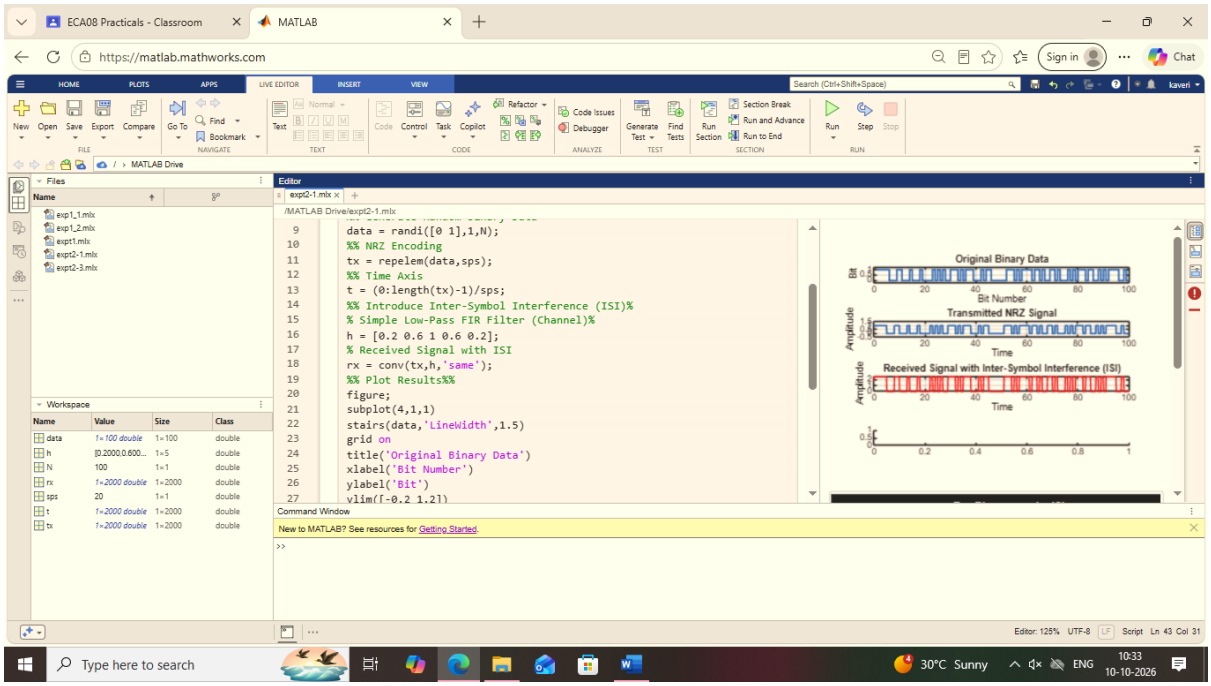
MATLAB SIMULATION AND EVALUATION OF EYE DIAGRAM PARAMETERS UNDER INTER-SYMBOL INTERFERENCE IN DIGITAL COMMUNICATION SYSTEMS

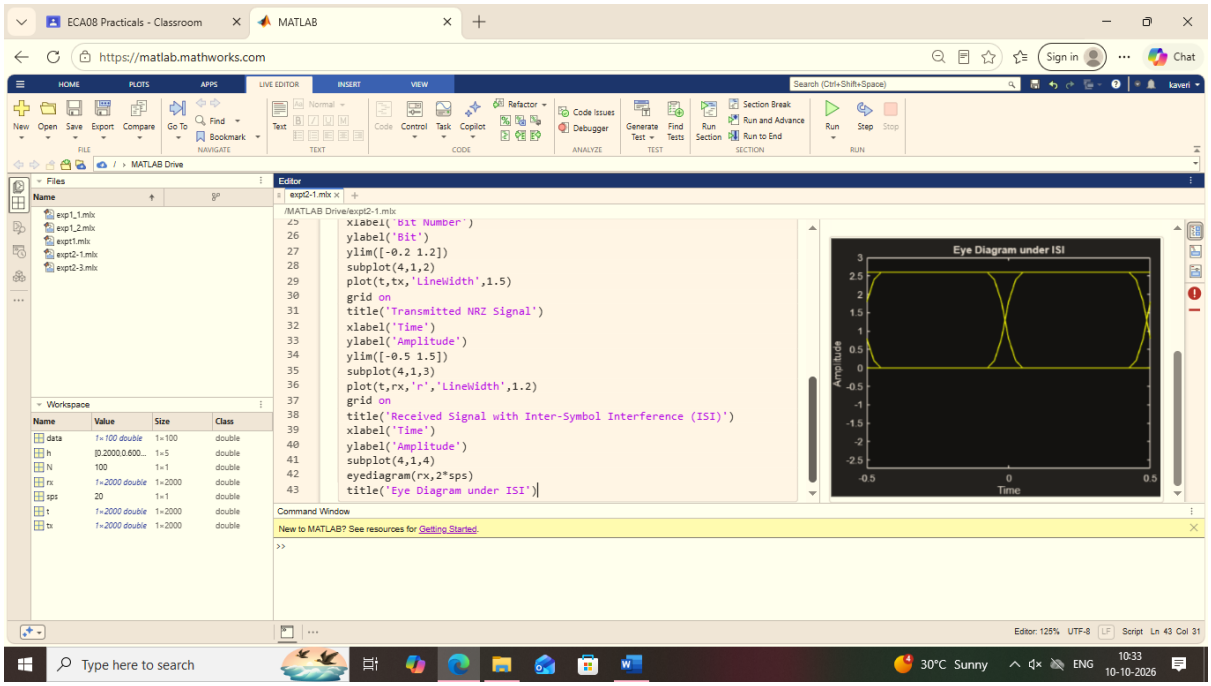
PROGRAM:

```
clc;
clear;
close all;
%% Parameters
N = 100; % Number of bits
sps = 20; % Samples per symbol
%% Generate Random Binary Data
data = randi([0 1],1,N);
%% NRZ Encoding
tx = repelem(data,sps);
%% Time Axis
t = (0:length(tx)-1)/sps;
%% Introduce Inter-Symbol Interference (ISI)%
% Simple Low-Pass FIR Filter (Channel)%
h = [0.2 0.6 1 0.6 0.2];
% Received Signal with ISI
rx = conv(tx,h,'same');
%% Plot Results%%
figure;
subplot(4,1,1)
stairs(data,'LineWidth',1.5)
grid on
title('Original Binary Data')
```

```
xlabel('Bit Number')
ylabel('Bit')
ylim([-0.2 1.2])
subplot(4,1,2)
plot(t,tx,'LineWidth',1.5)
grid on
title('Transmitted NRZ Signal')
xlabel('Time')
ylabel('Amplitude')
ylim([-0.5 1.5])
subplot(4,1,3)
plot(t,rx,'r','LineWidth',1.2)
grid on
title('Received Signal with Inter-Symbol Interference (ISI)')
xlabel('Time')
ylabel('Amplitude')
subplot(4,1,4)
eyediagram(rx,2*sps)
title('Eye Diagram under ISI')
```

OUTPUT:


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Generate Test
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Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
MATLAB Drive
Files
Name
expt1_1.mlx
expt1_2.mlx
expt1.mlx
expt2-1.mlx
expt2-3.mlx
Workspace
Name
Value
Size
Class
data
h
N
rx
sps
t
tx
Editor
expt2-1.mlx
/MATLAB Drive/expt2-1.mlx
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tx = repelem(data,sps);
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grid on
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ylabel('Bit')
ylim([-0.2 1.2])
Original Binary Data
Bit Number
Transmitted NRZ Signal
Time
Received Signal with Inter-Symbol Interference (ISI)
Command Window
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xlabel('Bit Number')
ylabel('Bit')
ylim([-0.2 1.2])
subplot(4,1,2)
plot(t,tx,'LineWidth',1.5)
grid on
title('Transmitted NRZ Signal')
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Eye Diagram under ISI
Amplitude
Time
Command Window
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